# Design Your Playground!

Intro



**Description**:
Given the length and width in term of expressions, students will find the perimeter and area of different rectangles by distributive property and skills of simplifying expressions.

Standards:
7.EE-1
7.EE-3

©TruePBL. All rights reserved.

Hello folks! The school building manager has a wild idea: he wants to transform our schoolyard into a temporary playground wonderland for the holiday season. And guess what? He's on the hunt for he most epic playground plan from our class!

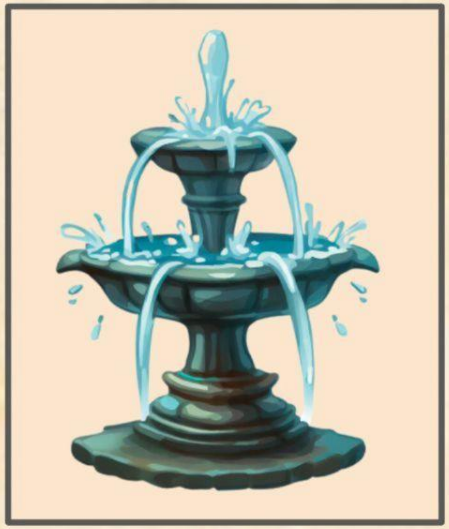
Hold on! He needs to interview you first. The next 6 slides are the interview questions. He only hires the person who gets correct answers!



©TruePBL. All rights reserved.

Need help?

# Fountain



$x+4$ yd

10 yd

What is the perimeter of this block?

$\square + \square + \square + \square$

$= \square$

What is the area of this block?

$\square \cdot \square$

$= \square$

©TruePBL. All rights reserved.

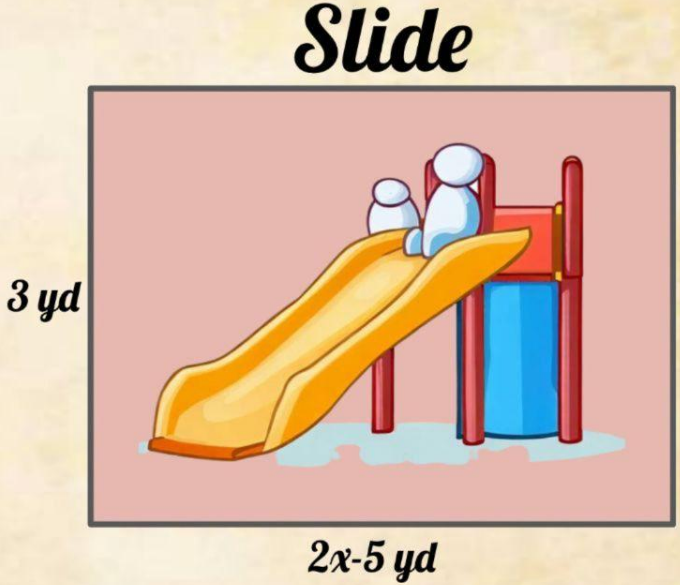
Need help?

What is the perimeter of this block?

$\square + \square + \square + \square = \square$

What is the area of this block?

$\square \cdot \square = \square$




©TruePBL. All rights reserved.

# Swing




What is the perimeter of this block?

□ + □ + □ + □ = □

What is the area of this block?

□ • □ = □

Need help?

©TruePBL. All rights reserved.

What is the perimeter of this block?

$\square + \square + \square + \square = \square$

What is the area of this block?

$\square \cdot \square = \square$

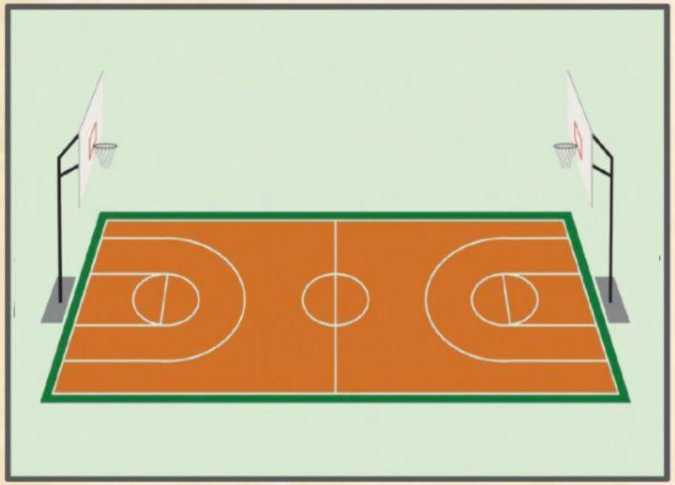
# Statue




Need help?

©TruePBL. All rights reserved.

# Basketball Court



2x yd

35 yd

What is the perimeter of this block?

[ ] + [ ] + [ ] + [ ]

= [ ]

What is the area of this block?

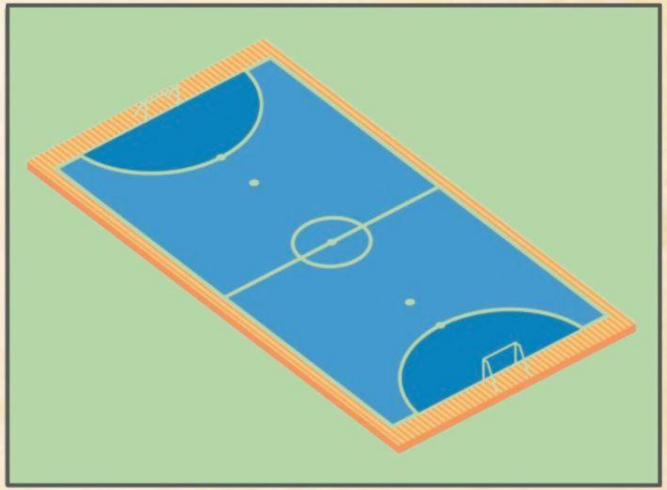
[ ] • [ ]

= [ ]

Need help?

©TruePBL. All rights reserved.

# Mini Soccer Field




What is the perimeter of this block?

□ + □ + □ + □

= □

What is the area of this block?

□ • □

= □

Need help?

©TruePBL. All rights reserved.



Done? Let's put the answers here!

Next up, the boss wants to offer you an advantage: how about your very own block in this playground?

©TruePBL. All rights reserved.

**Step 1:** Google any object/equipment that you want to put in the playground.








**Step 2:** Copy and paste the picture of it in the green box. Make sure it is inside the box.

**Step 3:** **Name you block!**

**Name:**

**Step 4:** What is the perimeter of this block?

___ + ___ + ___ + ___

= ___

What is the area of this block?

___ • ___

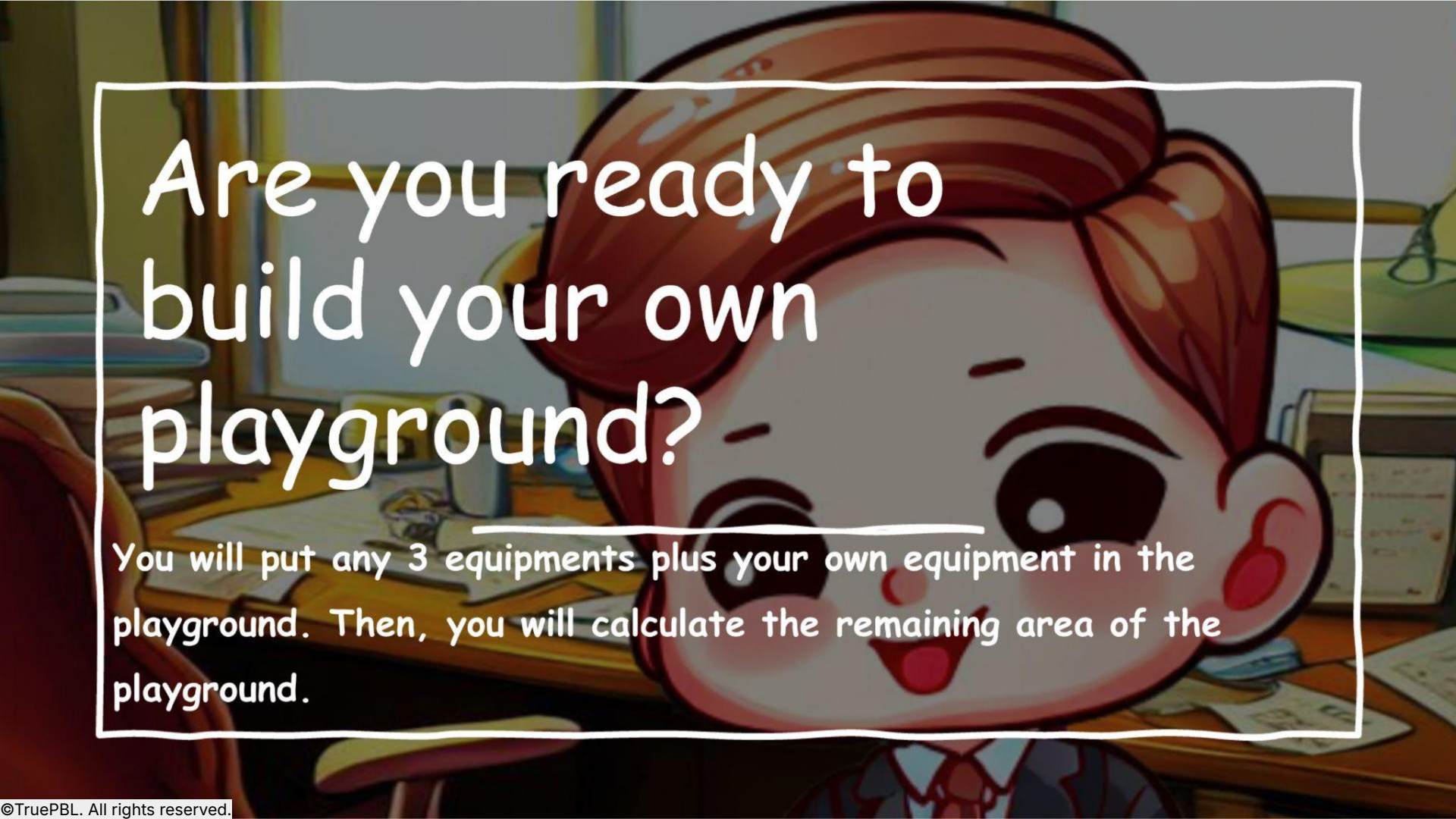
= ___

$2x+3y$ yd

Post the picture here!

23 yd

©TruePBL. All rights reserved.




©TruePBL. All rights reserved.

Copy and paste any 3 equipments into the next slide.

Then, post the picture of your own equipment in the next slide too!

$x+4$ yd



10 yd

2.5 yd



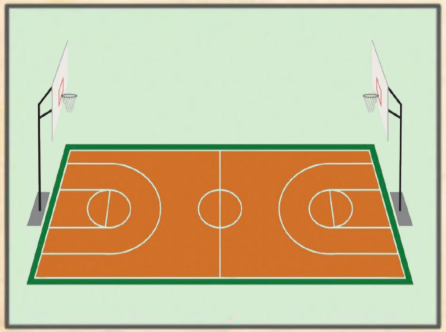
$x+2$ yd

3 yd



$2x-5$ yd

$2x$ yd



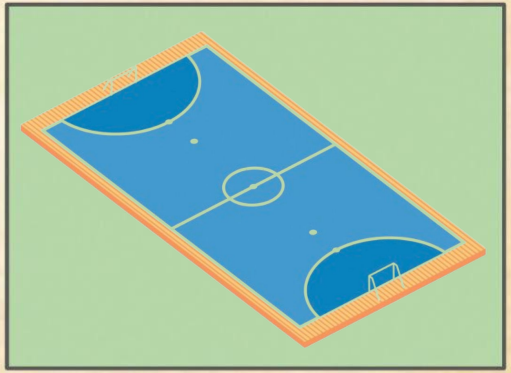
35 yd

$x-3$ yd



2 yd

28.2 yd



$4x + 6.4$ yd

©TruePBL. All rights reserved.




©TruePBL. All rights reserved.

Congratulations on reaching the final stage of the interview process. To your right, you will find the exact measurements of the playground. Your task is to calculate the area that remains uncovered, as this will help you determine how much of the playground needs to be covered with grass.

x+32 yd



400 yd

Area of this playground: (Show work)

Total area of your equipments: (Show work)

Remaining area for the grass: (Show work)

Need help?

©TruePBL. All rights reserved.